

MAS Michigan Residuals Land Application Operational Checklist

Facility: **Cadillac WWTP** Site: **21N08W30-AB01** Internal ID: **MI-MS-RI30-AB01**

Facility /Permit Criteria

Nicknam

1. Facility has an approved MDEQ Residuals Management Plan or AUA authorizing land application. ☒ FS
2. Residuals have been tested according to Permit and meet criteria for land application: ☒ TS
- a. Analytical up to date and meets criteria for land application ie <Table 1 Analysis Date **9/2/21** ☒ TS
- b. NANI has been executed ☒ TS
- c. CPLR notification if applicable sent ☒ TS
- d. Agreement form on file ☒ TS
- e. Permit allows for proposed application time (winter) & method IE Injection, Incorporation, Surface ☒ TS
3. Site is approved with all notices and waiting periods adhered to. MDEQ, Township, Health Department 10 days **AUA's Township and Health 21 days.**
- Date Land Application may begin on or after (RMP/AUA):** **11/25/19** **12/6/19** ☒ TS

Site Specific Criteria

1. Site recordkeeping information.
- a. Verify crop and yield, document restrictions of any crop with greater than 30 day harvest ☒ TS
- b. Soil test date is less than two years old - Date entered: **9/27/2021** ☒ TS
- c. Soil test P1<150 for RMP **and 75 ppm for AUA** Amount Entered **113 ppm** ☒ TS
- d. Application does not exceed N requirement of crop- Proposed N rate **138 lbs/a** ☒ TS
- e. If application will meet 90% of CPLR then notification has been sent ☒ TS
2. Site suitability criteria
- a. Water Table-Distance from soil surface to groundwater is > 30 inches ☐ SS
- b. Site field conditions conducive to land ap-IE not to wet, safe unloading conditions ☐ SS
- c. Site has been flagged or isolations otherwise marked IE-with applicator shanks ☐ SS
3. Site Application Information
- a. Application type: ☒ Injection ☐ Incorporation ☐ Surface

Injection Operator must maintain adequate soil coverage such that there is no "significant" biosolids on the surface 1 hour after application. There should be no pooling of biosolids or movement of biosolids in channel along slopes **regardless of applications rate.**

Incorporation of liquid or semi-solid/cake biosolids needs to be worked into the soil within 6 hours of Application.

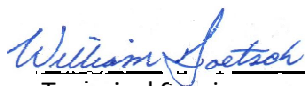
Surface application of any biosolids will require site specific instructions. **DO NOT APPLY** without this documentation.

b. Application Rate

Gallons/ Acre	Dry Tons/ Acre	Total Acres	Estimated % Solids	Total-Max Gallons
10,000	1.5	18	3.5%	180,000

c. Post Application Certification

Signing below certifies that all the above have been reviewed and the application meets MDEQ and Company standards. Please make any remarks that may improve our performance and public acceptance.

 **11/24/2021**
Technical Services Date

Site Supervisor Date

Conversion Sheet for Dry Tons At Differing % Solids

Facility: Cadillac WWTP

Site: MI-MS-RI30-AB01

Date:

11/24/21

%solids	Dry tons	Gal/A	Total Gallons
2.0%	1.5	17,500	315,000
2.5%	1.5	14,000	252,000
3.0%	1.5	11,667	210,000
3.5%	1.5	10,000	180,000
4.0%	1.5	8,750	157,500
4.5%	1.5	7,778	140,000
5.0%	1.5	7,000	126,000
5.5%	1.5	6,364	114,545
6.0%	1.5	5,833	105,000
6.5%	1.5	5,385	96,923
7.0%	1.5	5,000	90,000
7.5%	1.5	4,667	84,000
8.0%	1.5	4,375	78,750
8.5%	1.5	4,118	74,118
9.0%	1.5	3,889	70,000
9.5%	1.5	3,684	66,316

MAS Michigan Residuals Land Application Operational Checklist

Facility: Cadillac WWTP	Site: 21N08W30-AB02	Internal ID:	MI-MS-RI30-AB02
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Facility /Permit Criteria

Nicknam

1. Facility has an approved MDEQ Residuals Management Plan or AUA authorizing land application. ☒ FS
2. Residuals have been tested according to Permit and meet criteria for land application: ☒ TS
 - a. Analytical up to date and meets criteria for land application ie <Table 1 Analysis Date **9/2/21** ☒ TS
 - b. NANI has been executed ☒ TS
 - c. CPLR notification if applicable sent ☒ TS
 - d. Agreement form on file ☒ TS
 - e. Permit allows for proposed application time (winter) & method IE Injection, Incorporation, Surface ☒ TS
3. Site is approved with all notices and waiting periods adhered to. MDEQ, Township, Health Department 10 days **AUA's Township and Health 21 days.**

Date Land Application may begin on or after (RMP/AUA):

11/25/19

12/6/19 ☒ TS

Site Specific Criteria

1. Site recordkeeping information.
 - a. Verify crop and yield, document restrictions of any crop with greater than 30 day harvest ☒ TS
 - b. Soil test date is less than two years old - Date entered: **9/27/2021** ☒ TS
 - c. Soil test P1<150 for RMP **and 75 ppm for AUA** Amount Entered **124 ppm** ☒ TS
 - d. Application does not exceed N requirement of crop- Proposed N rate **158 lbs/a** ☒ TS
 - e. If application will meet 90% of CPLR then notification has been sent ☒ TS
2. Site suitability criteria
 - a. Water Table-Distance from soil surface to groundwater is > 30 inches ☐ SS
 - b. Site field conditions conducive to land ap-IE not to wet, safe unloading conditions ☐ SS
 - c. Site has been flagged or isolations otherwise marked IE-with applicator shanks ☐ SS

3. Site Application Information

a. Application type: ☒ Injection ☐ Incorporation ☐ Surface

Injection Operator must maintain adequate soil coverage such that there is no "significant" biosolids on the surface 1 hour after application. There should be no pooling of biosolids or movement of biosolids in channel along slopes **regardless of applications rate.**

Incorporation of liquid or semi-solid/cake biosolids needs to be worked into the soil within 6 hours of Application.

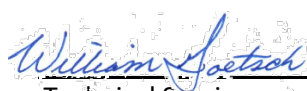
Surface application of any biosolids will require site specific instructions. **DO NOT APPLY** without this documentation.

b. Application Rate

Gallons/ Acre	Dry Tons/ Acre	Total Acres	Estimated % Solids	Total-Max Gallons
10,000	1.7	43	3.5%	430,000

c. Post Application Certification

Signing below certifies that all the above have been reviewed and the application meets MDEQ and Company standards. Please make any remarks that may improve our performance and public acceptance.

 **11/24/2021**
Technical Services Date

Site Supervisor Date

Conversion Sheet for Dry Tons At Differing % Solids

Facility: Cadillac WWTP

Site: MI-MS-RI30-AB02

Date:

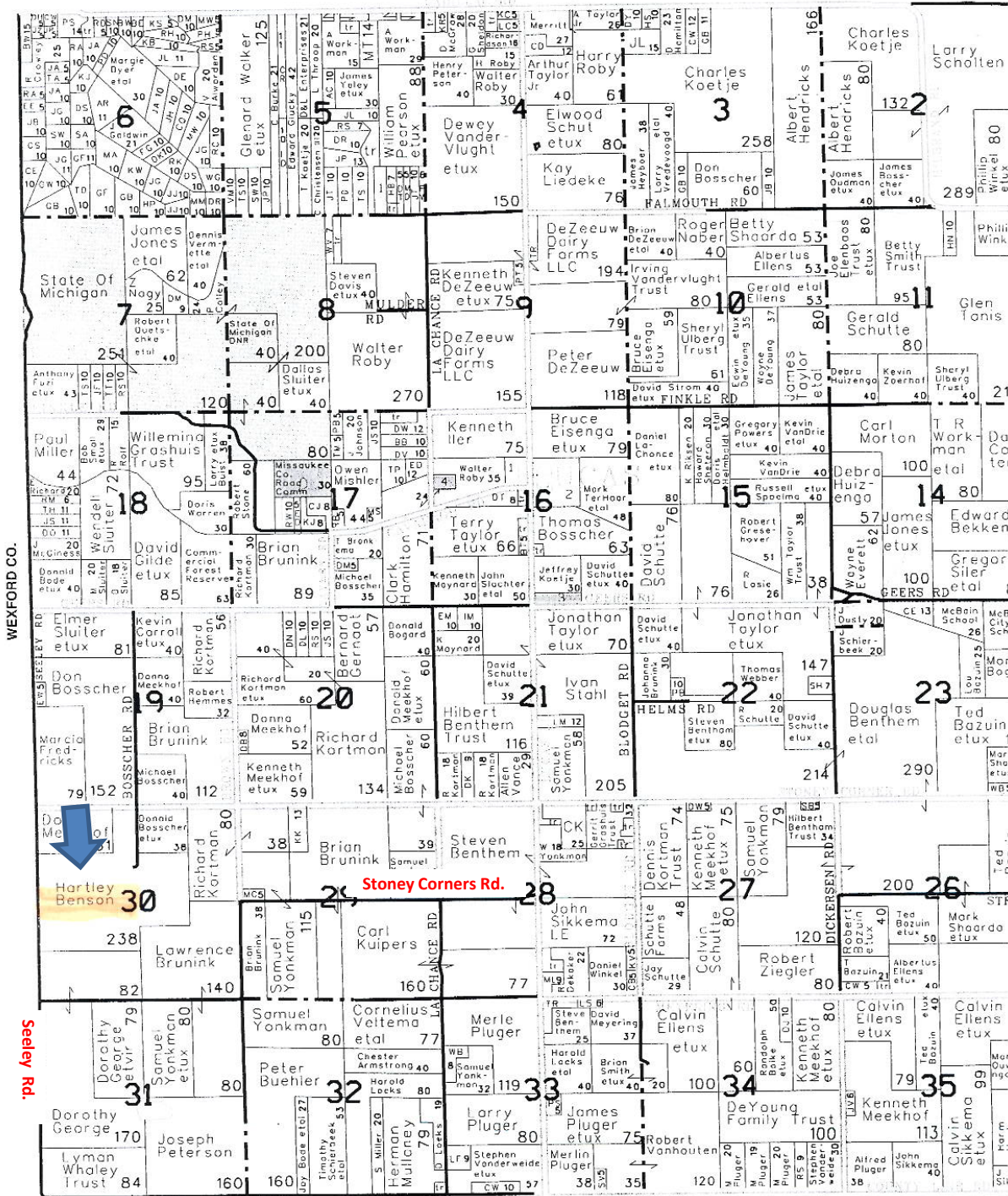
11/24/21

%solids	Dry tons	Gal/A	Total Gallons
2.0%	1.7	17,500	752,500
2.5%	1.7	14,000	602,000
3.0%	1.7	11,667	501,667
3.5%	1.7	10,000	430,000
4.0%	1.7	8,750	376,250
4.5%	1.7	7,778	334,444
5.0%	1.7	7,000	301,000
5.5%	1.7	6,364	273,636
6.0%	1.7	5,833	250,833
6.5%	1.7	5,385	231,538
7.0%	1.7	5,000	215,000
7.5%	1.7	4,667	200,667
8.0%	1.7	4,375	188,125
8.5%	1.7	4,118	177,059
9.0%	1.7	3,889	167,222
9.5%	1.7	3,684	158,421

T-21-N**RICHLAND PLAT**

(Landowners)

LAKE TWP. PAGE 30

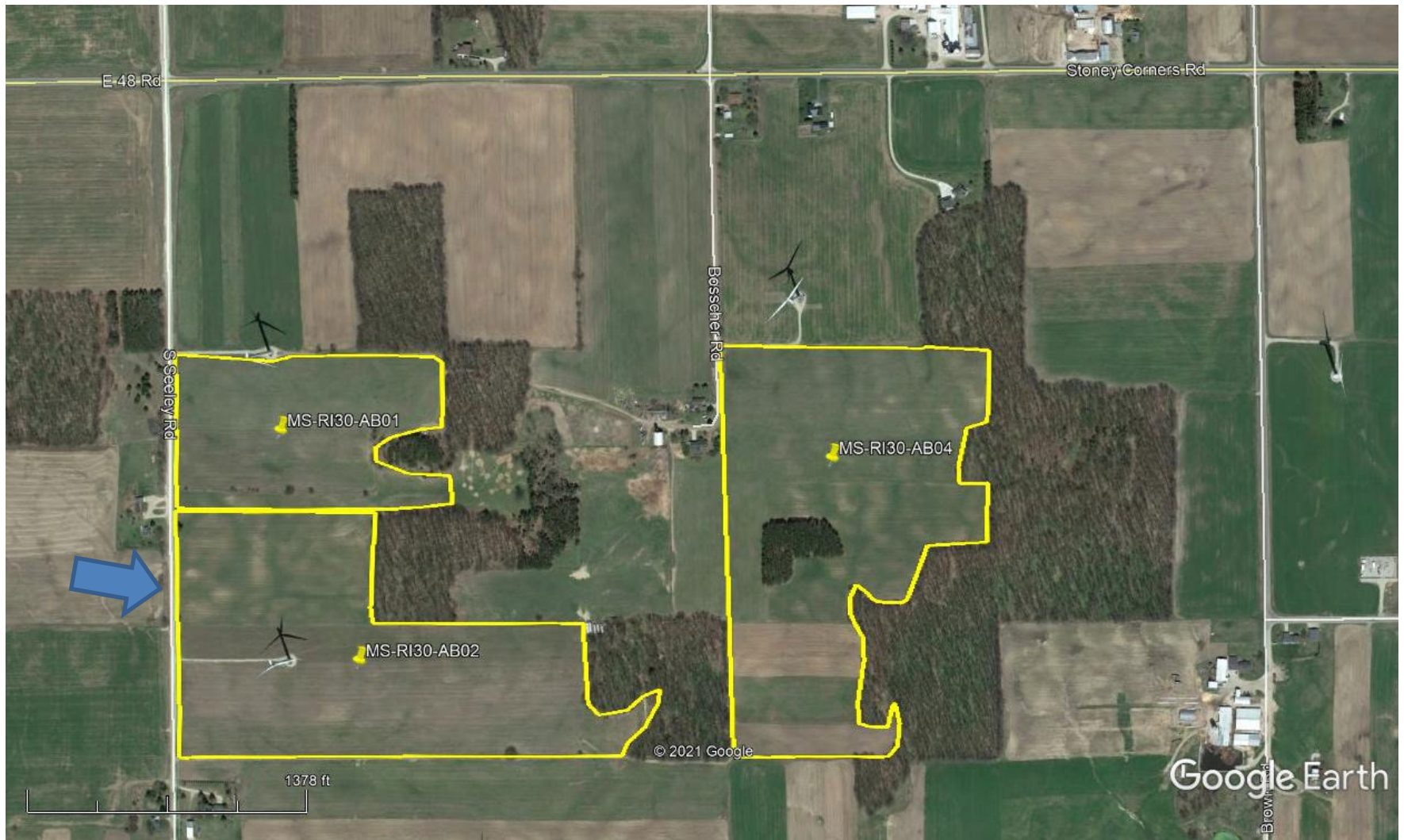


See Page 58 For Additional Names.

OSCEOLA CO.

Owner: Allen Benson
Operator: Mike Bosscher
Address: 11320 W. Stoney Corners Rd.
Cadillac, MI 49601
Phone: 231-499-4256
Mobile 231-499-4256
Site Comments: -

MDEQ ID: 21N08W30-AB02
Nickname: MI-MS-RI30-AB02
County: Missaukee
Township: Richland
Section: 30
Acres: 44.8



Owner: Allen Benson
Operator: Mike Bosscher
Address: 11320 W. Stoney Corners Rd.
Cadillac, MI 49601
Phone: 231-499-4256
Mobile 231-499-4256
Site Comments: -

MDEQ ID: 21N08W30-AB02
Nickname: MI-MS-RI30-AB02
County: Missaukee
Township: Richland
Section: 30
Acres: 44.8

MAS Michigan Residuals Land Application Operational Checklist

Facility: **Cadillac WWTP** Site: **21N08W30-AB03** Internal ID: **MI-MS-RI30-AB03**

Facility /Permit Criteria

Nicknam

1. Facility has an approved MDEQ Residuals Management Plan or AUA authorizing land application. ☒ FS
2. Residuals have been tested according to Permit and meet criteria for land application: ☒ TS
- a. Analytical up to date and meets criteria for land application ie <Table 1 Analysis Date **9/2/21** ☒ TS
- b. NANI has been executed ☒ TS
- c. CPLR notification if applicable sent ☒ TS
- d. Agreement form on file ☒ TS
- e. Permit allows for proposed application time (winter) & method IE Injection, Incorporation, Surface ☒ TS
3. Site is approved with all notices and waiting periods adhered to. MDEQ, Township, Health Department 10 days **AUA's Township and Health 21 days.**
- Date Land Application may begin on or after (RMP/AUA):** **11/25/19** **12/6/19** ☒ TS

Site Specific Criteria

1. Site recordkeeping information.
- a. Verify crop and yield, document restrictions of any crop with greater than 30 day harvest ☒ TS
- b. Soil test date is less than two years old - Date entered: **9/27/2021** ☒ TS
- c. Soil test P1<150 for RMP **and 75 ppm for AUA** Amount Entered **145 ppm** ☒ TS
- d. Application does not exceed N requirement of crop- Proposed N rate **129 lbs/a** ☒ TS
- e. If application will meet 90% of CPLR then notification has been sent ☒ TS
2. Site suitability criteria
- a. Water Table-Distance from soil surface to groundwater is > 30 inches ☐ SS
- b. Site field conditions conducive to land ap-IE not to wet, safe unloading conditions ☐ SS
- c. Site has been flagged or isolations otherwise marked IE-with applicator shanks ☐ SS

3. Site Application Information

a. Application type: ☒ Injection ☐ Incorporation ☐ Surface

Injection Operator must maintain adequate soil coverage such that there is no "significant" biosolids on the surface 1 hour after application. There should be no pooling of biosolids or movement of biosolids in channel along slopes **regardless of applications rate.**

Incorporation of liquid or semi-solid/cake biosolids needs to be worked into the soil within 6 hours of Application.

Surface application of any biosolids will require site specific instructions. **DO NOT APPLY** without this documentation.

b. Application Rate

Gallons/ Acre	Dry Tons/ Acre	Total Acres	Estimated % Solids	Total-Max Gallons
10,000	1.5	32	3.5%	320,000

c. Post Application Certification

Signing below certifies that all the above have been reviewed and the application meets MDEQ and Company standards. Please make any remarks that may improve our performance and public acceptance.

 **11/24/2021**
Technical Services Date

Site Supervisor Date

Conversion Sheet for Dry Tons At Differing % Solids

Facility: Cadillac WWTP

Site: MI-MS-RI30-AB03

Date:

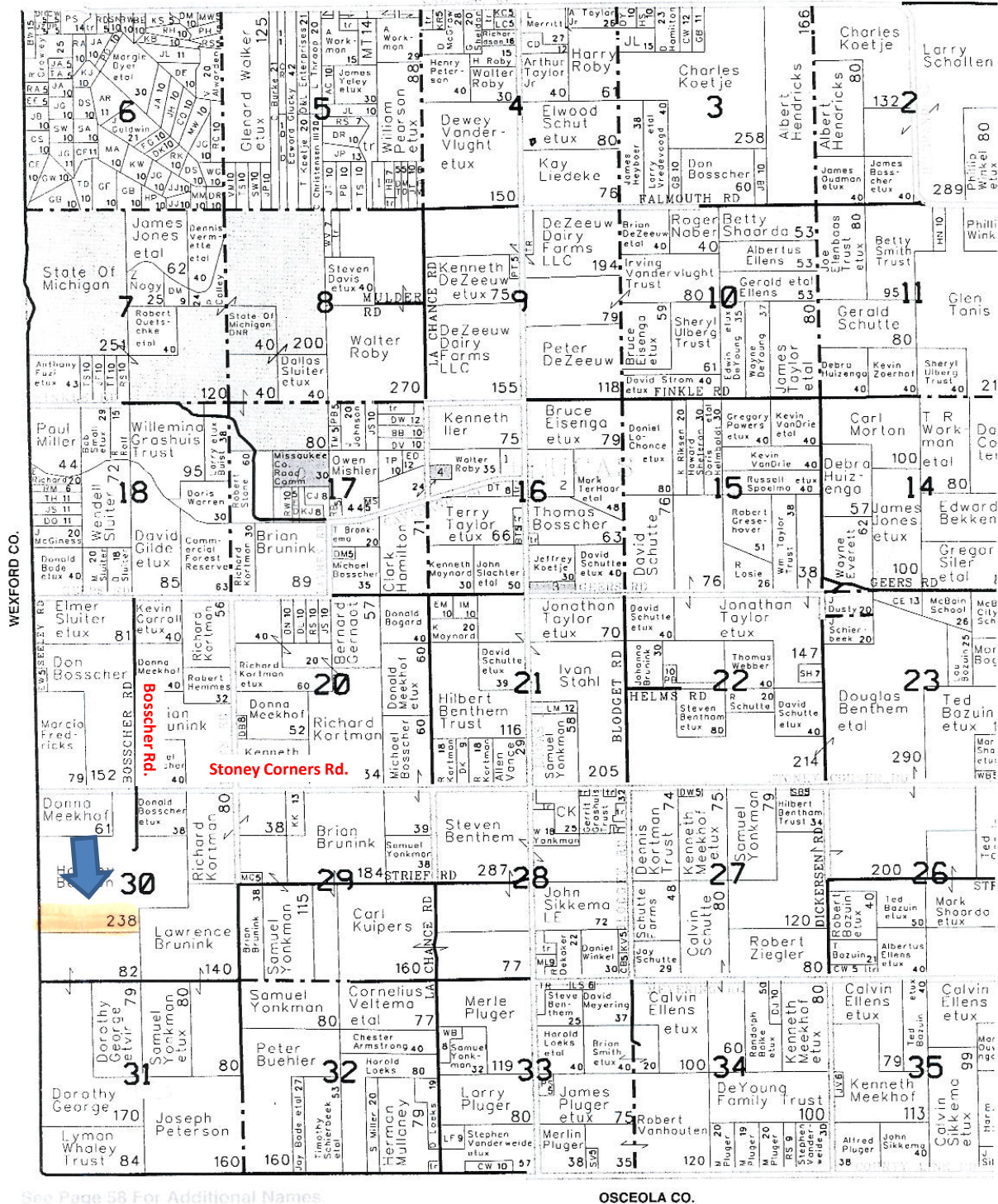
11/24/21

%solids	Dry tons	Gal/A	Total Gallons
2.0%	1.5	17,500	560,000
2.5%	1.5	14,000	448,000
3.0%	1.5	11,667	373,333
3.5%	1.5	10,000	320,000
4.0%	1.5	8,750	280,000
4.5%	1.5	7,778	248,889
5.0%	1.5	7,000	224,000
5.5%	1.5	6,364	203,636
6.0%	1.5	5,833	186,667
6.5%	1.5	5,385	172,308
7.0%	1.5	5,000	160,000
7.5%	1.5	4,667	149,333
8.0%	1.5	4,375	140,000
8.5%	1.5	4,118	131,765
9.0%	1.5	3,889	124,444
9.5%	1.5	3,684	117,895

T-21-N**RICHLAND PLAT**

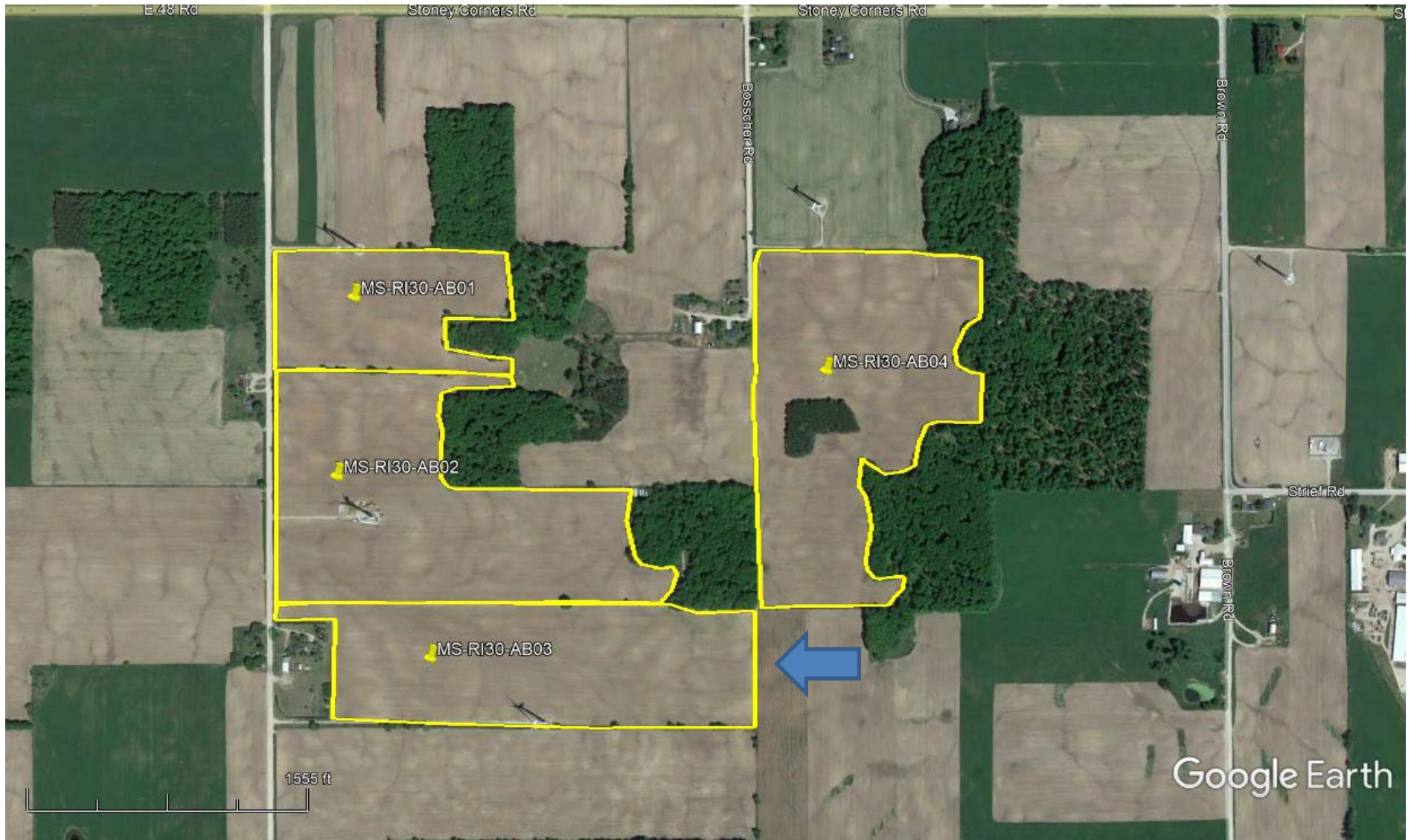
(Landowners)

LAKE TWP. PAGE 30



Owner: Allen Benson
Operator: Mike Bosscher
Address: 11320 W. Stoney Corners Rd.
Cadillac, MI 49601
Phone: 231-499-4256
Mobile 231-499-4256
Site Comments: -

MDEQ ID: 21N08W30-AB03
Nickname: MI-MS-RI30-AB03
County: Missaukee
Township: Richland
Section: 30
Acres: 35.6



Owner: Allen Benson
Operator: Mike Bosscher
Address: 11320 W. Stoney Corners Rd.
Cadillac, MI 49601
Phone: 231-499-4256
Mobile 231-499-4256
Site Comments: -

MDEQ ID: 21N08W30-AB03
Nickname: MI-MS-RI30-AB03
County: Missaukee
Township: Richland
Section: 30
Acres: 35.6



MAS Michigan Residuals Land Application Operational Checklist

Facility: **Cadillac WWTP** Site: **21N08W30-AB04** Internal ID: **MI-MS-RI30-AB04**

Facility /Permit Criteria

Nicknam

1. Facility has an approved MDEQ Residuals Management Plan or AUA authorizing land application. ☒ FS
2. Residuals have been tested according to Permit and meet criteria for land application: ☒ TS
- a. Analytical up to date and meets criteria for land application ie <Table 1 Analysis Date **9/2/21** ☒ TS
- b. NANI has been executed ☒ TS
- c. CPLR notification if applicable sent ☒ TS
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- e. Permit allows for proposed application time (winter) & method IE Injection, Incorporation, Surface ☒ TS
3. Site is approved with all notices and waiting periods adhered to. MDEQ, Township, Health Department 10 days **AUA's Township and Health 21 days.**
- Date Land Application may begin on or after (RMP/AUA):** **11/25/19** **12/6/19** ☒ TS

Site Specific Criteria

1. Site recordkeeping information.
- a. Verify crop and yield, document restrictions of any crop with grater than 30 day harvest ☒ TS
- b. Soil test date is less than two years old - Date entered: **9/27/2021** ☒ TS
- c. Soil test P1<150 for RMP **and 75 ppm for AUA** Amount Entered **132 ppm** ☒ TS
- d. Application does not exceed N requirement of crop- Proposed N rate **129 lbs/a** ☒ TS
- e. If application will meet 90% of CPLR then notification has been sent ☒ TS
2. Site suitability criteria
- a. Water Table-Distance from soil surface to groundwater is > 30 inches ☐ SS
- b. Site field conditions conducive to land ap-IE not to wet, safe unloading conditions ☐ SS
- c. Site has been flagged or isolations otherwise marked IE-with applicator shanks ☐ SS

3. Site Application Information

a. Application type: ☒ Injection ☐ Incorporation ☐ Surface

Injection Operator must maintain adequate soil coverage such that there is no "significant" biosolids on the surface 1 hour after application. There should be no pooling of biosolids or movement of biosolids in channel along slopes **regardless of applications rate.**

Incorporation of liquid or semi-solid/cake biosolids needs to be worked into the soil within 6 hours of Application.

Surface application of any biosolids will require site specific instructions. **DO NOT APPLY** without this documentation.

b. Application Rate

Gallons/ Acre	Dry Tons/ Acre	Total Acres	Estimated % Solids	Total-Max Gallons
10,000	1.5	42	3.5%	420,000

c. Post Application Certification

Signing below certifies that all the above have been reviewed and the application meets MDEQ and Company standards. Please make any remarks that may improve our performance and public acceptance.

 **11/24/2021**
Technical Services Date

Site Supervisor Date

Conversion Sheet for Dry Tons At Differing % Solids

Facility: Cadillac WWTP

Site: MI-MS-RI30-AB04

Date:

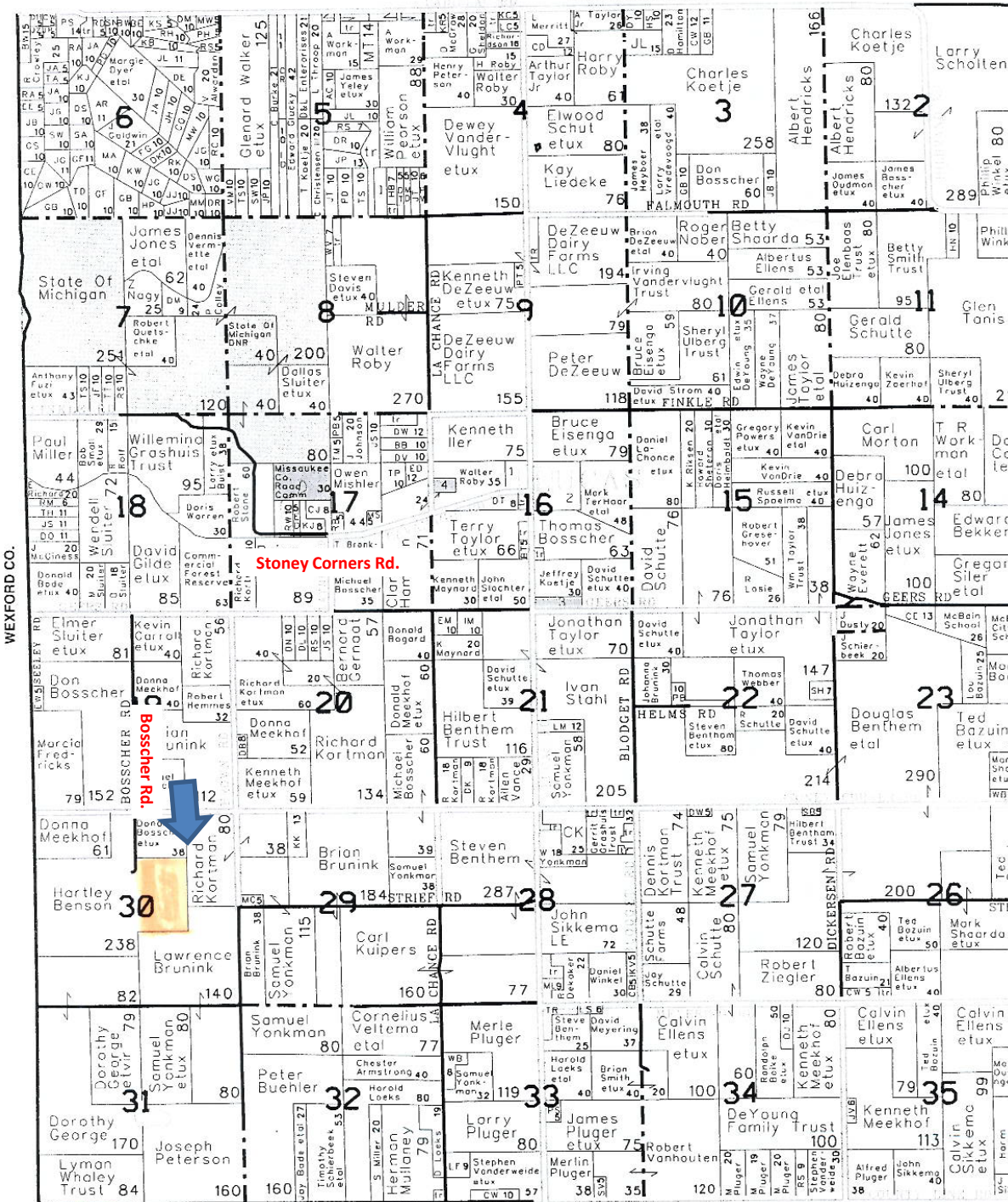
11/24/21

%solids	Dry tons	Gal/A	Total Gallons
2.0%	1.5	17,500	735,000
2.5%	1.5	14,000	588,000
3.0%	1.5	11,667	490,000
3.5%	1.5	10,000	420,000
4.0%	1.5	8,750	367,500
4.5%	1.5	7,778	326,667
5.0%	1.5	7,000	294,000
5.5%	1.5	6,364	267,273
6.0%	1.5	5,833	245,000
6.5%	1.5	5,385	226,154
7.0%	1.5	5,000	210,000
7.5%	1.5	4,667	196,000
8.0%	1.5	4,375	183,750
8.5%	1.5	4,118	172,941
9.0%	1.5	3,889	163,333
9.5%	1.5	3,684	154,737

T-21-N**RICHLAND PLAT**

(Landowners)

LAKE TWP. PAGE 30

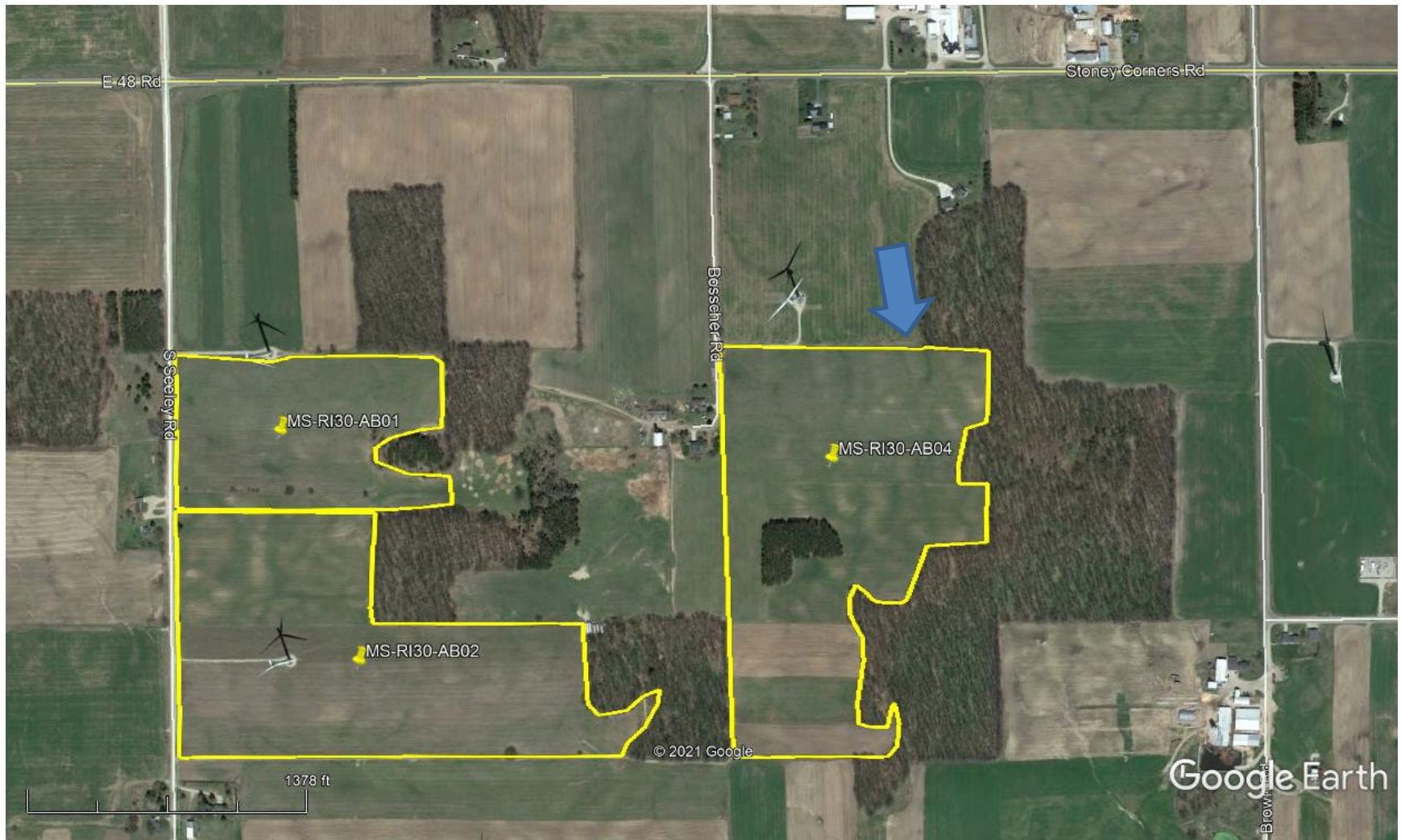


See Page 58 For Additional Names.

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